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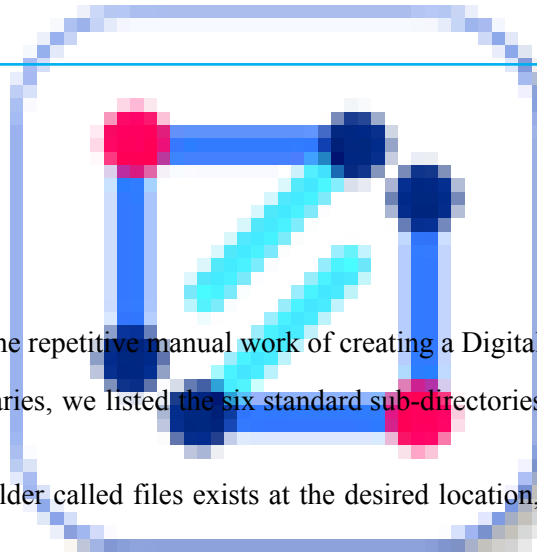
1. In-Lab Tasks:

i. Task 3:

The goal of Task 3 was to remove the repetitive manual work of creating a Digital IC project's folder tree.

Using Python's `os` and `pathlib` libraries, we listed the six standard sub-directories—`rtl`, `tb`, `scripts`, `docs`, `synthesis`, and `simulation`.

The script first ensures a parent folder called `files` exists at the desired location, then loops through the list to create each child directory.



```
• [cc@ncdc-0063 src]$ python3 task_3.py

Base directory: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files

../files/rtl created successfully!
../files/tb created successfully!
../files/scripts created successfully!
../files/docs created successfully!
../files/synthesis created successfully!
../files/simulation created successfully!

• [cc@ncdc-0063 src]$ python3 task_3.py

Base directory: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files

../files/rtl is already exists
../files/tb is already exists
../files/scripts is already exists
../files/docs is already exists
../files/synthesis is already exists
../files/simulation is already exists

○ [cc@ncdc-0063 src]$ █
```

A key part of the method is `Path.mkdir(exist_ok=True)`.

This call creates the directory only if it is missing and silently skips it if it already exists.

While doing so, the script prints a status message: *created successfully* when a folder is new, or *already exists* when the folder is present.

This guarantees a consistent directory layout every time the script runs and prevents accidental overwrites, making it ideal for setting up multiple design projects.

Scripting

ii. Task 4:

Task 4 builds on the structure created in Task 3 by automatically generating a **SystemVerilog module template** inside the `rtl` folder.

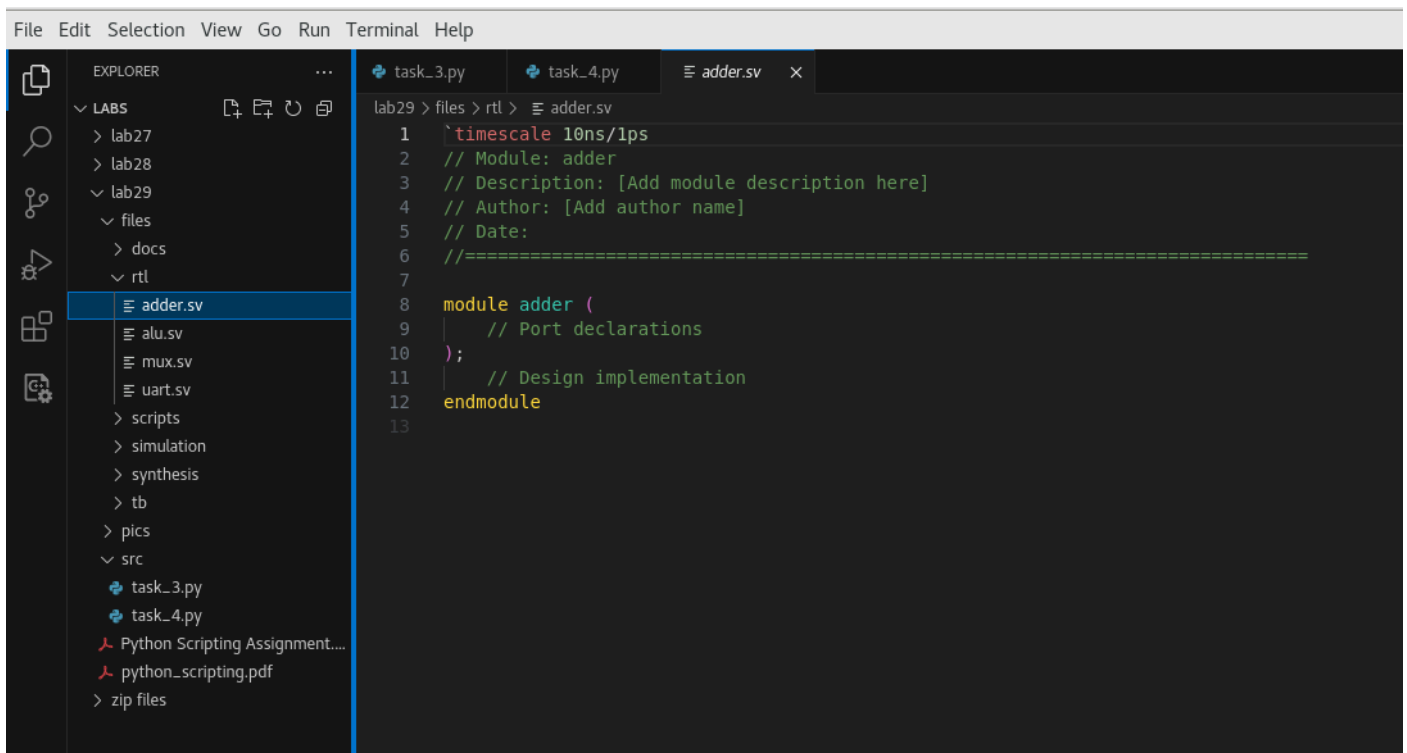
The script accepts the module name as a command-line argument, for example `python task_4_module_template.py alu`

It calculates the correct path to `files/rtl`, creates the folder hierarchy if it is missing (`parents=True`, `exist_ok=True`), and then writes a new `<module_name>.sv` file.

```
[cc@ncdc-0063 src]$ python3 task_4.py adder
[OK] Created: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files/rtl/adder.sv
[cc@ncdc-0063 src]$ python3 task_4.py uart
[OK] Created: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files/rtl/uart.sv
[cc@ncdc-0063 src]$ python3 task_4.py alu
[OK] Created: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files/rtl/alu.sv
[cc@ncdc-0063 src]$ python3 task_4.py alu
[WARNING] /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files/rtl/alu.sv already exists. Aborting to avoid overwrite.
[cc@ncdc-0063 src]$ python3 task_4.py mux
[OK] Created: /home/cc/muddassir_ali_siddiqui/ncdc/scripting/labs/lab29/files/rtl/mux.sv
[cc@ncdc-0063 src]$
```

Inside the file, a simple skeleton is inserted: a timescale directive, header comments, the `module` declaration with the provided name, and an empty port list.

Before writing, the script checks whether the file already exists to avoid overwriting existing code, warning the user if it does.



This automation ensures that every new design module starts from a consistent, documented template and that it is always stored in the correct RTL directory defined by Task 3.

2. Critical Analysis:

Lab 29 demonstrated how Python can automate routine setup tasks in digital IC design. In Task 3 we scripted the creation of a complete project hierarchy—`files/rtl`, `tb`, `scripts`, `docs`, `synthesis`, and `simulation`—while reporting whether each folder was newly made or already present. Task 4 then generated a ready-to-edit SystemVerilog module inside the `rtl` folder, creating any missing directories and avoiding overwrites. Together these steps remove repetitive manual work, reduce errors, and ensure every project starts from a clean, consistent structure, showcasing Python's ability to streamline hardware-design workflows.